

Problem

Our goal is to create a GUI that can convert a value in seconds into hours, minutes, and seconds.



a) Main Scenario:

The user will enter a value between 0 and 86,400 seconds. When the 'Convert' button is pressed, the system will calculate the equivalent time in hours, minutes, and seconds. The input is represented by 's', and the outputs are 'ho' (hours), 'min' (minutes), and 'sec' (seconds).

The formula used is as follows: $s = 3600 * ho + 60 * min + sec$.

b) Exception Scenarios:

1. EmptyFieldExce

إخفاء

إيقاف المشاركة

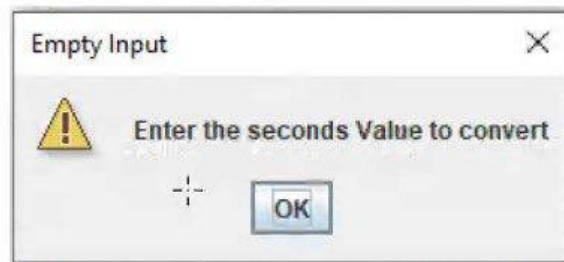
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مشارك شاشتك

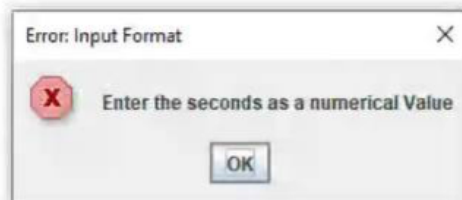
lank.

b) Exception Scenarios:

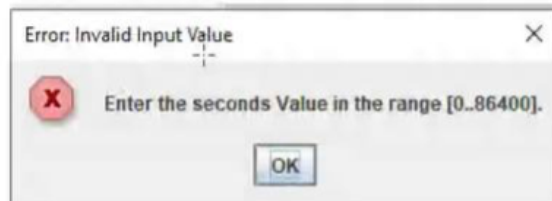
1. EmptyFieldException: Thrown when the input JTextField is left blank.



2. NumberFormatException: Thrown when the input in the JTextField cannot be parsed as a number.



3. InvalidSecondsException: Thrown when the input value falls outside the allowed range.



Note: All the exception classes are already defined.

Consider the following exampl

إخفاء

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- **Input:** 12,500 seconds
- **Result:** 3 hours, 28 minutes, and 20 seconds.

Let's verify:

$$3 * 3600 + 28 * 60 + 20 = 12,500 \text{ seconds.}$$

Steps to Implement:

1. Create a class called TimeConverter: (5 Marks)

- Define the class as the main GUI window.

2. Define all GUI components as attributes: (3 Marks)

- The GUI components like buttons, text fields, and labels should be defined as instance variables (attributes) within the TimeConverter class.

3. Handle exceptions with JOptionPane: (6 Marks)

- Use JOptionPane.showMessageDialog to display error messages when handling exceptions, such as EmptyFieldException, NumberFormatException, and InvalidSecondsException.

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4. Use inner classes BConvert and BCancel to handle events: (6 Marks)

- Implement event listeners using inner classes for handling button clicks (BConvert for conversion and BCancel for clearing).

5. Set the GUI size to 450x100:

- Use the setSize(450, 100) method in the constructor to set the size of the window.

I Appendix: Components and Methods

Component / Method	Description	Methods Used
JFrame	A top-level container for the GUI window.	setSize(), setDefaultCloseOperation(), setVisible(), setLayout().
JTextField	Allows the user to input text.	getText(), setText().
JLabel	Displays static text.	

هذا الجدول زيادة كإضافة
من كتاب الورد والفاصل
من مشيل من مشيل